Appendix C

Format for Thermodynamic Data Coefficients

This appendix explains the format for data contained in the file thermo.inp (app. D). Equations (1) to (3) are repeated here for convenience:

$$C_p^o(T)/R = a_1T^{-2} + a_2T^{-1} + a_3 + a_4T + a_5T^2 + a_6T^3 + a_7T^4 \qquad (1)$$

$$H^o(T)/RT = -a_1T^{-2} + a_2 lnT/T + a_3 + a_4T/2 + a_5T^2/3 + a_6T^3/4 + a_7T^4/5 + b_1/T \qquad (2)$$

$$S^o(T)/R = -a_1T^{-2}/2 - a_2T^{-1} + a_3 lnT + a_4T + a_5T^2/2 + a_6T^3/3 + a_7T^4/4 + b_2 \qquad (3)$$

TABLE C1.—FORTRAN FORMAT USED FOR DATA IN APPENDIX D

Record	Contents	FORTRAN format	Columns
1	Species name or formula	A16	1 to 16
	Comments and data sources	A62	19 to 80
2	Number of T intervals	I2	1 to 2
	Reference date code	A6	4 to 9
	Chemical formula—symbols (all capitals) and numbers	5(A2, F6.2)	11 to 50
	Zero for gas; nonzero for condensed[a]	I2	51 to 52
	Molecular weight	F13.7	53 to 65
	Heat of formation at 298.15 K, J/mol	F15.5	66 to 80
3	Temperature range	2F11.3	1 to 22
	Number of coefficients for $C_p^o(T)/R$ (always seven)	I1	23
	T exponents in empirical equation for $C_p^o(T)/R$ [always –2, –1, 0, 1, 2, 3, 4; see eq. (1)]	8F5.1	24 to 63
	$H^o(298.15) - H^o(0)$ J/mol, if available	F15.3	66 to 80
4	First five coefficients for $C_p^o(T)/R$, eq. (1)	5D16.9	1 to 80
5	Last two coefficients for $C_p^o(T)/R$, eq. (1)	2D16.9	1 to 32
	Integration constants b_1 and b_2, eqs. (2) and (3)	2D16.9	49 to 80
-	Repeat 3, 4, and 5 for each interval	---------	----------

[a] Condensed phases are numbered in increasing order by temperature.

For example, the following data are for condensed titanium nitride:

```
12345678901234567890123456789012345678901234567890123456789012345678901234567890
         10        20        30        40        50        60        70        80

TiN(cr)           Chase,1998 pp1612-4.
 2 j 6/68 TI  1.00N   1.00    0.00    0.00    0.00 1     61.87374    -337648.800
    200.000    800.0007 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0            5487.000
-5.479117220D+05 9.328691110D+03-6.386263890D+01 2.429925456D-01-4.304234520D-04
 3.792645100D-07-1.317412256D-10                -8.424256140D+04 3.392988560D+02
    800.000   3220.0007 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0            5487.000
-3.656247060D+05 1.265730431D+03 3.831711190D+00 1.632900455D-03-1.062786626D-07
 1.310931390D-11-5.770548410D-16                -5.027654400D+04-1.652632899D+01
TiN(L)            Chase,1998 pp1612-4.
 1 j 6/68 TI  1.00N   1.00    0.00    0.00    0.00 2     61.87374    -337648.800
   3220.000   6000.0007 -2.0 -1.0  0.0  1.0  2.0  3.0  4.0  0.0            5487.000
 0.000000000D+00 0.000000000D+00 7.548249987D+00 0.000000000D+00 0.000000000D+00
 0.000000000D+00 0.000000000D+00                -3.626039860D+04-3.958296649D+01

12345678901234567890123456789012345678901234567890123456789012345678901234567890
         10        20        30        40        50        60        70        80
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